



How to Use DeepSeek



Natasha Ong

<think>

To determine the number of 'r's in "strawberry," I'll start by examining each letter in the word.

First, I'll list out the letters: S, T, R, A, W, B, E, R, R, Y.

Next, I'll identify and count all the lowercase 'r's. Scanning through, I find three 'r's: the second letter, and two more towards the end.

Therefore, there are a total of three 'r's in "strawberry."

</think>

To determine how many ****r****s are in the word **"strawberry"**, let's analyze each letter:

1. **S**trawberry
2. **T**
3. **R**
4. **A**
5. **W**
6. **B**
7. **E**
8. **R**
9. **R**
10. **Y**

Now, let's count the number of times the letter 'r' appears:

- The second letter is '**R**'.
- The eighth letter is '**R**'.
- The ninth letter is '**R**'.

Total 'r's: 3





Introducing Today's Project!

In this project, we will demonstrate how to use DeepSeek and make the most of this LLM that seems to be performing at the same level as OpenAI - but at a much lower price point. We're doing this project to learn whether DeepSeek is the new fav LLM.

Tools and concepts

Services we used were DeepSeek, Ollama, Chatbox and the OpenAI API platform. Key concepts we learnt include advanced reasoning, self hosting, token efficiency and temperature settings. After reviewing DeepSeek vs OpenAI, we personally preferred ...

Project reflection

This project took me approximately 2.5 hours to complete, including demo time. The most challenging part was DeepSeek's overthinking. It was most rewarding to compare token efficiency and using Chatbox for a visual experience with DeepSeek locally.

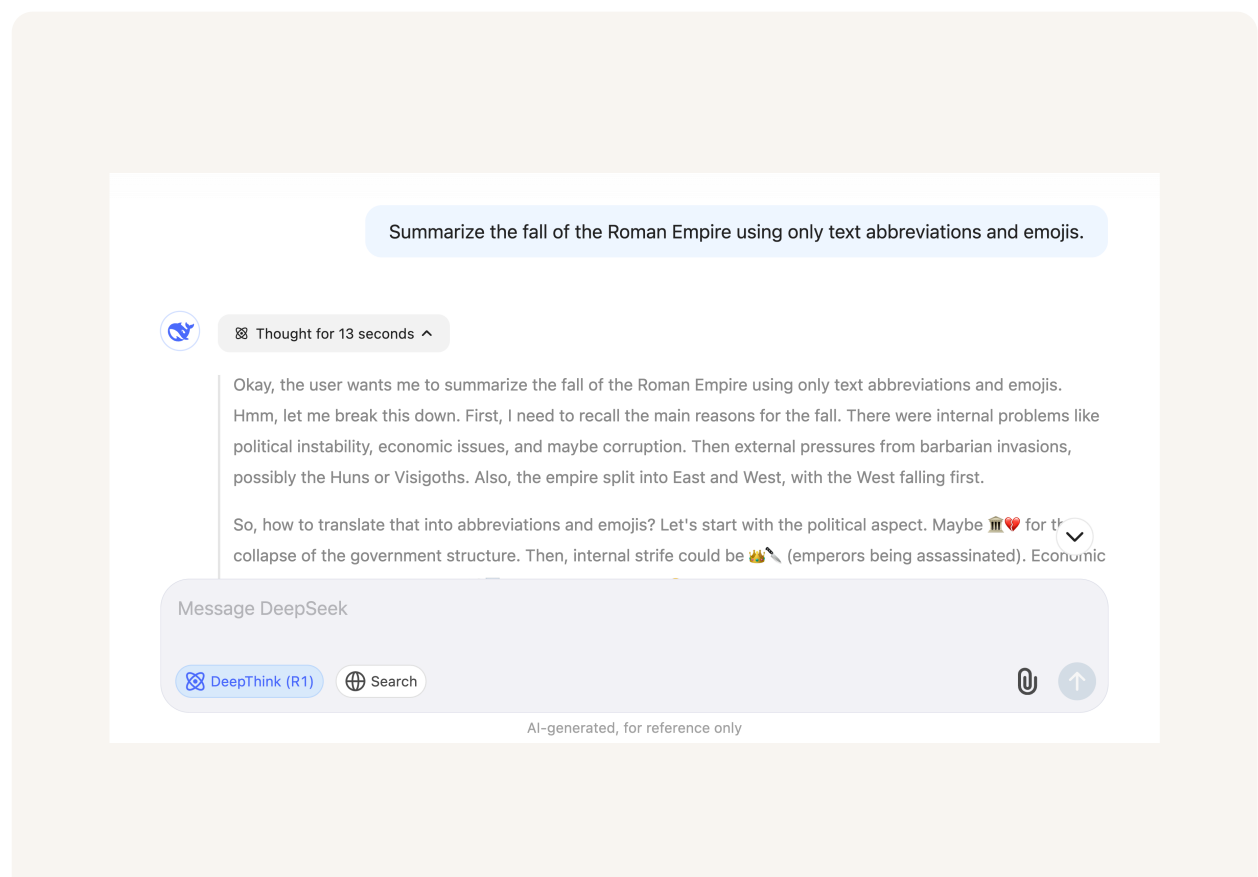
We did this project today to better understand DeepSeek and form our own opinions around whether it is something we want to switch to compared to OpenAI / other popular LLMs. This project has a hands on focus and helped us see both the pros and cons.



Exploring DeepSeek

DeepSeek is a company that develops Large Language Models (LLMs). Their R1 model gained attention for performing at the same level as OpenAI's latest o1 model - but is also open source. We have access to a world-class LLM for free/low price.

While you could access DeepSeek over the web app, some concerns are privacy (i.e. where is data stored), constant internet connection needed (i.e. connection gets cut off without the internet) and latency (i.e slow response times during high traffic)





Ollama and DeepSeek R1

Ollama is a software for using LLMs locally on our computer. It is helpful because it manages the downloading, installing and updating of an LLM for us, which usually would take a lot of manual work.

You won't be able to find OpenAI models in Ollama because OpenAI models are not open source. This means the configuration, the codebase and the architecture that makes up OpenAI models are confidential and not available in open-source LLM platforms.

We tested using DeepSeek offline by turning Wifi off and running commands over the terminal and observed that we are still getting responses back from DeepSeek. You can also see 'think' tags in the terminal that represent DeepSeek's thinking process.

```
>>> I'm so excited to talk to you over the terminal
<think>
Okay, the user just said they're really excited to talk about
it. Maybe they want to set up a chat or something.

I should invite them to start talking soon.
</think>

Great! Feel free to start the conversation with me anytime. 😊
```



DeepSeek R1 Sizes

DeepSeek R1 has different model sizes, which comes in varying levels of processing needs and accuracy. This is helpful for running DeepSeek locally because we can pick and choose which model fits our hardware requirements best.

The R1 model you choose to run locally depends on the memory and storage space in your computer. We chose the 8b model (i.e. 8 billion parameters) because it offers a substantial improvement from the 1.5b model and is still accessible for our computer

Settings

MODEL DISPLAY CHAT ADVANCED

Model Provider:
OLLAMA API

API Host
http://127.0.0.1:11434 RESET

Please ensure that the Remote Ollama Service is able to connect remotely. For more details, refer to [this tutorial](#).

Model
deepseek-r1:1.5b

Max Message Count in Context
20

Temperature
0.8

Meticulous Creative

CANCEL SAVE



Chatbox

To complete our local setup, we installed Chatbox to have a visual interface as we chat with DeepSeek R1 locally. Our Chatbox settings use Ollama API as the API model (i.e. the tool that will connect us with the desired LLM) and the 1.5b R1 model.

We tested two different R1 model sizes, which were 1.5b and 8b using the prompt 'How many r's are in strawberry?' The results made it clear that 1.5b was the lighter model - it responded with only 2 r's in "strawberry", whereas the 8b model said 3.

```
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1. Strawberry
2. T
3. R
4. A
5. W
6. B
7. E
8. R
9. R
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Total 'r's: 3
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Temperature Settings

The temperature setting in an LLM determines how creative and 'out of the box' it will be when it generates a response. To see this in action, we dialled up the temperature to the maximum (2) and prompted DeepSeek to generate a recipe.

We started a third chat with ChatGPT to act as a judge for the responses - can ChatGPT tell which one was generated with a high temperature? ChatGPT's analysis will also help us with understanding how to spot low vs high temperature generated text.

ChatGPT quickly figured out which piece was generated with a high temperature, because it identified that the high temperature piece featured more creative ingredients, more steps and more complexity. The low temperature response is also simpler.

 The first response was generated with a higher temperature setting. Here's why:

- Creativity & Complexity** – The first response includes more complex and gourmet-style elements, such as pistachio garnish, grated chocolate, gelatin for texture, and a ganache layer. High-temperature settings often produce more creative and diverse outputs, incorporating unexpected elements.
- Structured & Detailed** – The first response has multiple layers and steps, including a separate ganache and mousse preparation, which adds a refined and sophisticated feel. Higher temperatures tend to introduce more details and complexity.
- More Ingredients & Variations** – The first response lists optional variations like vanilla extract or citrus zest and includes different textures (whipped cream, gelatin, nuts, and salt crystals). This suggests a more experimental and free-flowing generation.
- Second Response is More Conservative** – The second response is simpler, more structured, and to the point, which is characteristic of a lower-temperature setting. It follows a straightforward approach, sticking to essential ingredients and methods without as much variation.
- Surprising Element in the Second Response** – The toast component in the second recipe feels a bit out of place. This might be a sign of randomness, but it's less elaborate than the layered textures and multiple elements in the first response.

Conclusion:

The first response was generated at a higher temperature due to its greater complexity.

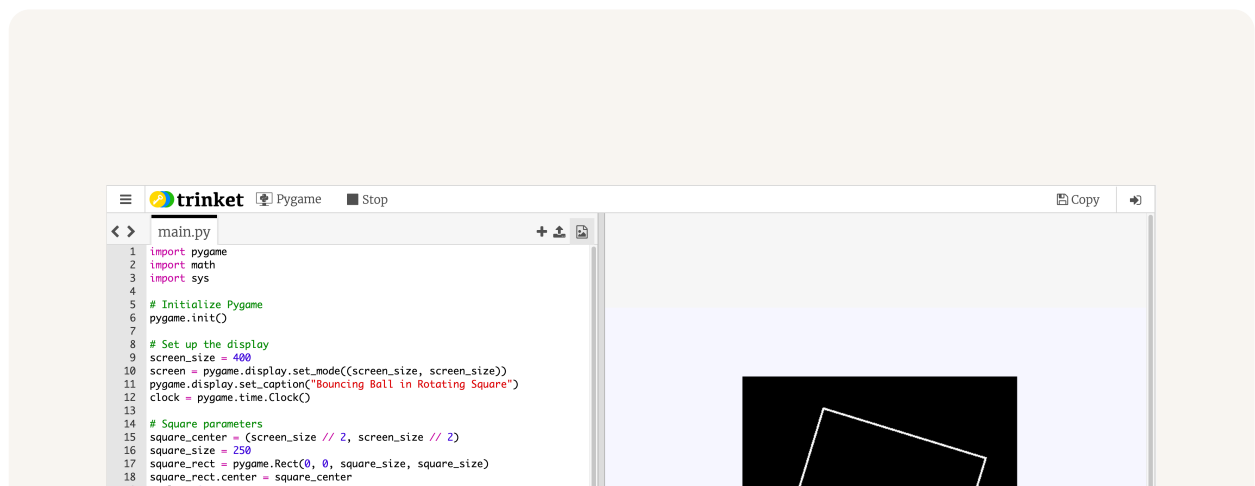


DeepSeek vs. ChatGPT Reasoning

We decided to compare DeepSeek R1 with OpenAI by prompting both to generate a Python script that animates a rotating square with a bouncing yellow ball inside. This is a challenging prompt because it involves rotation mathematics + Python animation.

To test ChatGPT's response, we put ChatGPT's python script into Trinket and watched it run live. ChatGPT's results came instantly, but the code itself did not run successfully - there was a rotating square, but no bouncing yellow ball inside :(

Compared to ChatGPT's performance, we thought DeepSeek's response was more accurate - the code it produced rendered a functioning model of what we wanted - although it took three minutes to generate. We prioritize accuracy so we preferred DeepSeek.





DeepSeek vs OpenAI Tokens

In a project extension, we're also comparing OpenAI and DeepSeek's token efficiency. We could access OpenAI's API by setting up a brand new account and accessing a test key.

When we used a higher temperature, OpenAI's model's response became incoherent - totally irrelevant to the prompt. This is likely because of a different training style and treatment of '2.0' as a measure of creativity/predictability of the response

Token efficiency refers to how many (or little) tokens an LLM uses/processes in order to generate a response. For the same request, DeepSeek used ~300 tokens whereas OpenAI's model used ~500 tokens. For developers, this means DeepSeek is cheaper.

This recipe offers a rich, dessert-like experience by combining subtle saltiness of sea salt.

tokens used: 325, model: Ollama (deepseek-r1:8b)